SUPPORTING ADULT GUIDANCE

Proof and problem-solving

Circle Theorems · Lesson 10. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This closing lesson turns from calculation to proof. Learners assemble the angle-at-the-centre proof by ordering statement/reason cards (a step out of order, using an unestablished fact, is rejected), practise supplying the reason for each line of an argument, and watch a second theorem — the cyclic quadrilateral — proved on top of the first. The aim is rigour: an ordered chain of justified statements, and a clear sense of why a proof is stronger than any number of measured examples.

Driving Question: *What turns a set of true statements into a watertight proof?*

The mathematics, in plain terms

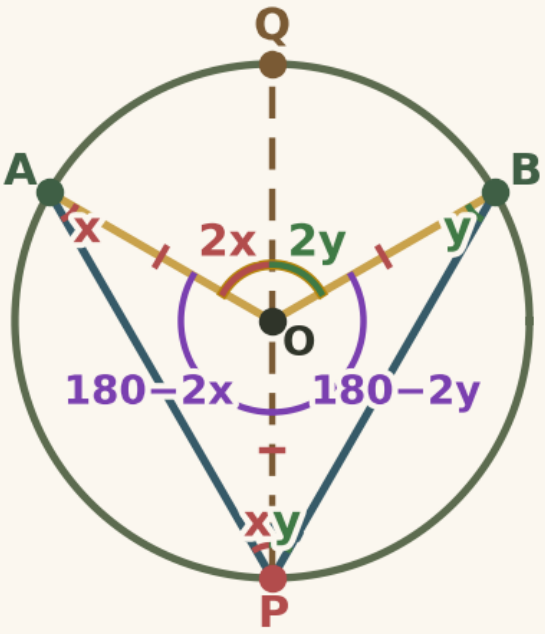
The keystone proof: extend PO to Q; OA = OB = OP give isosceles triangles with base angles x and y; the exterior-angle fact gives ∠AOQ = 2x, ∠BOQ = 2y; hence ∠AOB = 2(x+y) = 2∠APB. The cyclic-quadrilateral proof applies this twice: with ∠ABC = b and ∠ADC = d, the two central angles on the opposite arcs are 2b and 2d, and since they make a full turn, 2b + 2d = 360°, so b + d = 180°. Order and reasons are the whole point.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Prove the angle at the centre theorem

P is on the circle; A and B are two more points; O is the centre. Prove that ∠AOB = 2∠APB.

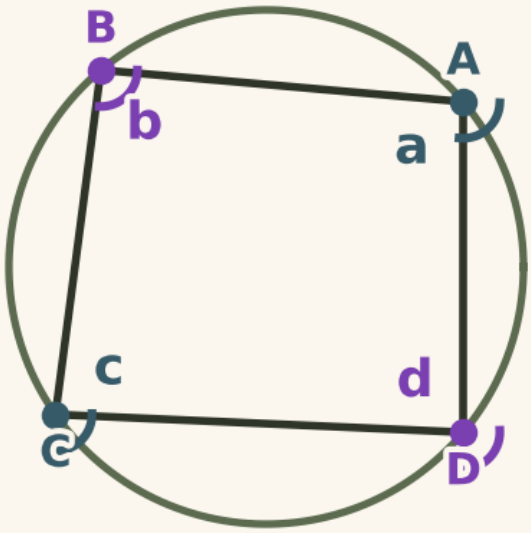


1. Join PO and extend it to Q. OA = OB = OP (radii), so triangles OAP and OBP are isosceles.	all radii are equal
2. ∠OAP = ∠OPA = x and ∠OBP = ∠OPB = y.	base angles of isosceles triangles
3. ∠AOQ = $2x$ and ∠BOQ = $2y$.	exterior angle = sum of the two opposite interior angles
4. ∠AOB = ∠AOQ + ∠BOQ = $2x + 2y = 2(x + y)$, and ∠APB = $x + y$.	adding the angles

∠AOB = 2∠APB — the angle at the centre is twice the angle at the circumference. ■

Example 2 — Prove the cyclic quadrilateral theorem

ABCD is a cyclic quadrilateral. Prove that ∠ABC + ∠ADC = 180°, using the theorem just proved.



1. Let ∠ABC = b and ∠ADC = d. Draw radii OA and OC.	set up the angles
2. ∠AOC = 2b (angle at the centre, standing on arc ADC with ∠ABC).	angle at the centre = 2 × angle at circumference
3. The reflex angle AOC = 2d (standing on arc ABC with ∠ADC).	angle at the centre, again
4. 2b + 2d = 360° (angles around O), so b + d = 180°.	angles around a point

∠ABC + ∠ADC = 180° — opposite angles of a cyclic quadrilateral are supplementary. ■

Order and reasons are the proof

Insist on two things: a valid order (no step may use a fact not yet established) and a reason for every statement. The proof builder enforces the first; your questioning should enforce the second. Draw out the idea that a proved result becomes a tool — the cyclic-quadrilateral proof reuses the keystone. Distinguish clearly between measuring (checks one case) and proving (settles all cases); the Studio task deliberately measures the keystone as a check on, not a substitute for, the written proof. This closes the Circle Theorems unit.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Lay the foundations of the keystone proof in order.	~9 min
Movement	Supply the reason for each statement.	~10 min
Reflection	Build the full six-step angle-at-the-centre proof.	~14 min
Creativity	Watch the cyclic-quadrilateral theorem proved on the keystone.	~9 min
Rest	A short pause on what a proof really is.	~2 min
Nutrition	Intellectual: constructing an unarguable case.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Using a statement before it is justified.
'Can you use that yet? Every step must follow from what you have already established.'

Giving statements with no reasons.
'A statement without a reason is not proof — name the fact that makes each line true.'

Believing a measured example proves the general case.
'Measuring one diagram checks one case. What would make it certain for every circle?'

Losing the thread of a long proof.
'What have you established so far, and what does it let you do next? Keep the chain in order.'

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: all the circle theorems, isosceles base angles, the exterior-angle fact, and that angles around a point add to 360°. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner asserts a step with no reason, or thinks a measurement is a proof. The aim is a question that restores rigour.

- “What is your reason for that line? Name the fact.”
- “Have you established everything this step depends on?”
- “Does measuring it once prove it always? What would?”
- “Which proved theorem could you use as a tool here?”

If they stall, that is the moment for a Reconnection Route or the lesson's prompt and hint — not for the answer.

Questions you might have

What is the difference between an angle chase and a proof?

A chase finds particular values; a proof shows a result holds in general, with every step justified. A proof usually uses letters (x, y) rather than specific numbers.

Why does the order of steps matter?

Each step may only use facts already established. Using a statement before you have justified it breaks the chain — the argument is no longer watertight.

Is a measured example ever enough?

No. Measuring confirms a single case and can catch mistakes, but only a proof shows a result is true for every possible circle.

Why prove things we already believe are true?

Because believing is not knowing. A proof shows precisely why a result must hold, reveals how the theorems connect, and lets each proved result safely support the next.

What is a 'show that' problem?

One that ends in a statement you must justify rather than a number to compute. You work towards the required statement, giving a reason at each step, and finish by stating that it now follows.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.

Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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